SSLC

2025



PART

C

# BIOLOGY

*learning material.....*



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CHAPTER-5

# LIFE PROCESSES

## TOPIC-1 NUTRITION

All living things perform certain life processes like growth, excretion, respiration, circulation and reproduction, etc.

Energy required to carry out the different life processes, is obtained from carbon-based food sources through nutrition.

There are two modes of nutrition:

**(i) Autotrophic nutrition:** It is a kind of nutrition in which organism prepare its own food by a process called photosynthesis. For e.g., Green plants and some photosynthetic bacteria.

**(ii) Heterotrophic nutrition:** It is a kind of nutrition in which an organism takes food from another organism. For e.g., Animals and fungi. It is of three types: Holozoic (e.g. Amoeba, animals), Saprophytic (e.g., fungi) and Parasitic (e.g., Cuscuta, ticks and mites)

### Nutrition in Plants

Carbon and energy requirements of the autotrophic organism are fulfilled by photosynthesis. It is the process by which autotrophs take in substances from the outside and convert them into stored forms of energy. This material is taken in the form of carbon dioxide and water which is converted into carbohydrates in the presence of sunlight and chlorophyll.

$$6CO_2 + 12H_2O \xrightarrow[\text{Sunlight}]{\text{Chlorophyll}} \underset{\text{(Glucose)}}{C_6H_{12}O_6} + 6O_2 + 6H_2O$$

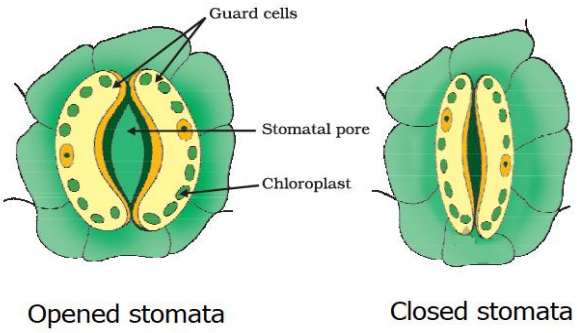
**Steps involved in photosynthesis :**

(i) Absorption of light energy by chlorophyll.
(ii) Conversion of light energy to chemical energy and splitting of water molecules into hydrogen and oxygen.
(iii) Reduction of carbon dioxide to carbohydrates.

## Exchange of gasses :

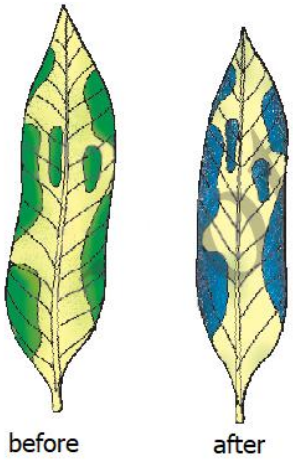
- Massive amounts of gaseous exchange takes place in the leaves through tiny pores called **Stomata** for the purpose of photosynthesis.
- The plant closes these pores when it does not need carbon dioxide for photosynthesis.
- The opening and closing of the pore is regulated by the guard cells.
- The guard cells swell when water flows into them, causing the stomatal pore to open. Similarly the pore closes if the guard cells shrink.

(Note : exchange of gases occurs across the surface of stems and roots as well.)



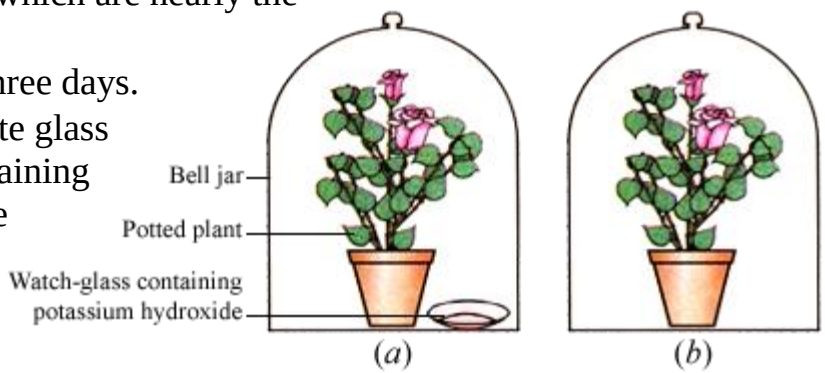

## Activity to demonstrates that chlorophyll is essential for photosynthesis.

- Take a potted plant with variegated leaves – for example, money plant or crotons.
- Keep the plant in a dark room for three days so that all the starch gets used up.
- Now keep the plant in sunlight for about six hours.
- Pluck a leaf from the plant. Mark the green areas in it and trace them on a sheet of paper.
- Test a leaf for the presence of starch
- Only the parts which were green previously, turn blue-black with iodine.
- This shows that chlorophyll is necessary for photosynthesis.




## Activity to demonstrates that $CO_2$ is essential for photosynthesis.

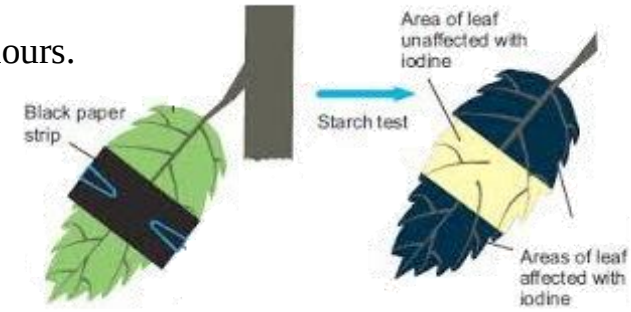
- Take two healthy potted plants which are nearly the same size.
- Keep them in a dark room for three days.
- Now place each plant on separate glass plates. Place a watch-glass containing potassium hydroxide by the side of one of the plants. The potassium hydroxide is used to absorb carbon dioxide.



- Cover both plants with separate bell-jars
- Use vaseline to seal the bottom of the jars to the glass plates so that the set-up is air-tight.
- Keep the plants in sunlight for about two hours.
- Pluck a leaf from each plant and check for the presence of starch
- Leaf plucked from the plant where potassium hydroxide was kept do not turn blue-black color with iodine
- This shows that chlorophyll is necessary for photosynthesis.

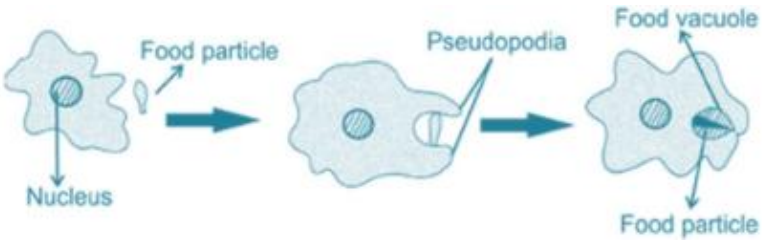
### Activity to demonstrates that sunlight is essential for photosynthesis.

- Take a potted plant
- Clip a black paper strip to the leaf
- Now keep the plant in sunlight for about six hours.
- Pluck a leaf from the plant and remove the black paper.
- Test a leaf for the presence of starch
- Only the parts which are exposed to sunlight are turn blue-black with iodine.
- This shows that sunlight is necessary for photosynthesis.



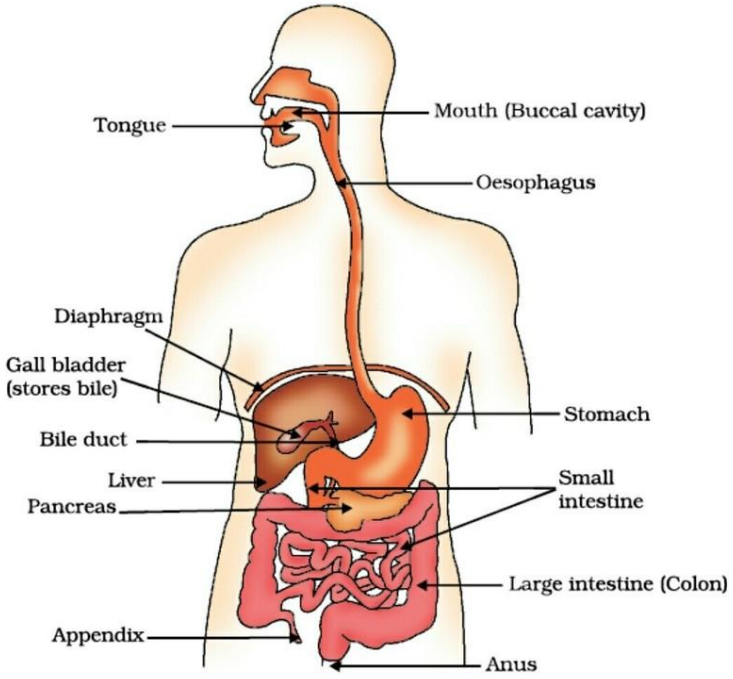

## Nutrition in Amoeba :

- Amoeba takes in food using temporary finger-like extensions called pseudopodia
- These extensions encircle over the food particle forming a food-vacuole.
- Inside the food vacuole, complex substances are broken down into simpler ones
- The digested food diffuse into the cytoplasm.
- The remaining undigested material is moved to the surface of the cell and thrown out.




## Nutrition in Human beings :

- The human digestive system comprises of alimentary canal and associated digestive glands.
- The alimentary canal is a long muscular tube extending from the mouth to the anus.
- Associated glands include salivary gland**,** gastric gland, Liver and Pancreas.




- In mouth , our teeth crushes the food in to smaller particles and of same texture.
- Saliva secreted by the salivary glands helps in the softening of food. The saliva contains an enzyme called **salivary amylase** that breaks down starch in to simple sugar.
- From the mouth, the food is taken to the stomach through the food-pipe or oesophagus. It was done by rhythmic movement of oesophagus. This is called **peristaltic movement.**

## Role of stomach/Gastric glands

- The digestion in stomach is taken care of by the gastric glands present in the wall of the stomach. These release hydrochloric acid, a protein digesting enzyme called pepsin, and mucus.
- The hydrochloric acid creates an acidic medium which facilitates the action of the enzyme pepsin. It kills microbes present in the food.
- The mucus protects the inner lining of the stomach from the action of the acid

From the stomach, the food now enters the small intestine. This is the longest part of the alimentary canal. The length of the small intestine differs in various animals depending on the food they eat.
**Herbivores** eating grass need a **longer small intestine** to allow the cellulose to be digested. Meat is easier to digest, hence **carnivores** have a **shorter small intestine**.

## Role of small intestine

- The small intestine is the site of the complete digestion of carbohydrates, proteins and fats.

- It receives the secretions of the liver and pancreas
- Pancreatic enzymes act in alkaline medium. The food coming from the stomach is acidic and has to be made alkaline. This is accomplished by the bile juice secreted from the liver.
- Fats are present in the intestine in the form of large globules. Bile salts break them down into smaller globules increasing the efficiency of enzyme action.
- The pancreas secretes pancreatic juice which contains enzymes like trypsin for digesting proteins and lipase for breaking down emulsified fats.
- The walls of the small intestine contain glands which secrete intestinal juice. The enzymes present in it finally convert the proteins to amino acids, complex carbohydrates into glucose and fats into fatty acids and glycerol.
- The inner lining of the small intestine has numerous finger-like projections called villi which increase the surface area for absorption. The villi are richly supplied with blood vessels which take the absorbed food

The unabsorbed food is sent into the large intestine where its wall absorb more water from this material. The rest of the material is removed from the body via the anus. The exit of this waste material is regulated by the anal sphincter.

### Digestive enzymes and their action :

| Enzyme | Source | Digestive Action |
|---|---|---|
| Salivary Amylase | Salivary Glands | Begins carbohydrate digestion by breaking down starch and glycogen to disaccharides |
| Lipase | Pancreas | Breaks down fats into fatty acids and glycerol |
| Trypsin | Pancreas | Breaks down proteins or partially digested proteins into peptides |
| Peptidase | Intestinal Enzymes | Breaks down peptides into amino acids |
| Sucrase, Maltase, Lactase | | Breaks down disaccharides into monosaccharides |
| Intestinal Lipase | | Breaks down fats into fatty acids and glycerol |

## TOPIC-2 RESPIRATION

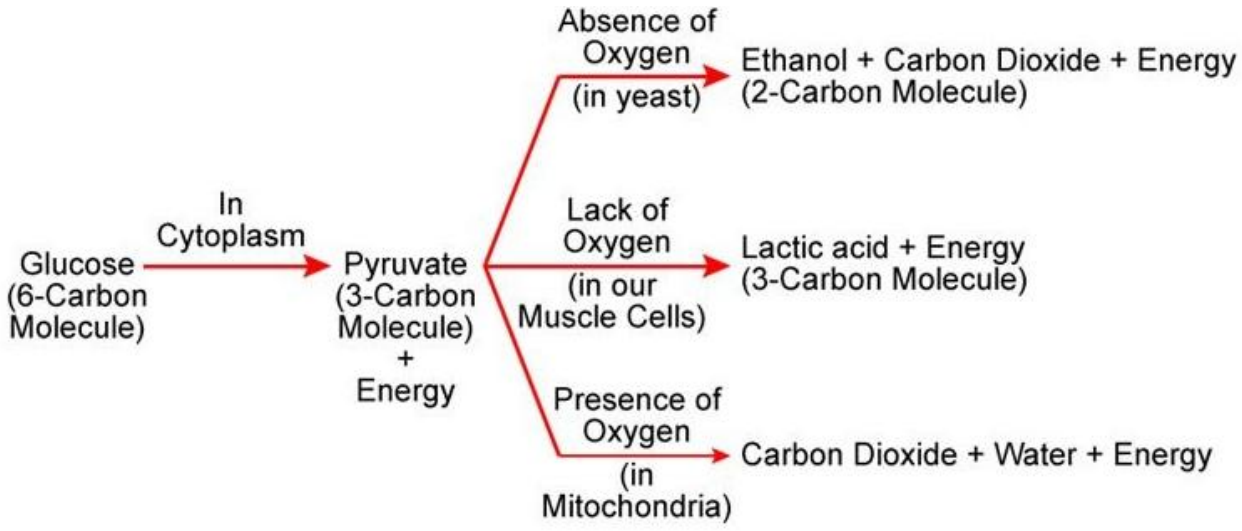
Respiration is the process in living organisms, which involves:

(i) Breathing (Gaseous exchange): Intake of oxygen from the atmosphere and release of $CO_2$.
(ii) Breakdown of simple food in order to release energy inside the cell.

## Types of Respiration :

| Aerobic | Anaerobic |
|---|---|
| Takes place in the presence of oxygen. | Takes place in the absence of oxygen. |
| Occurs in mitochondria. | Occurs in cytoplasm. |
| End products are $CO_2$ and H2O. | End products are alcohol and $CO_2$ |
| More amount of energy is released. | Less amount of energy is released. |
| Examples: Most plants and animals. | Examples: bacteria, yeast |

## Breakdown of Glucose by Various Pathways:




In plants exchange of gases carried through stomata (leaves), lenticels (stem) and roots. Carbon dioxide and oxygen are exchanged by the process called diffusion. The direction of diffusion depends upon the environmental conditions and the requirements of the plant.

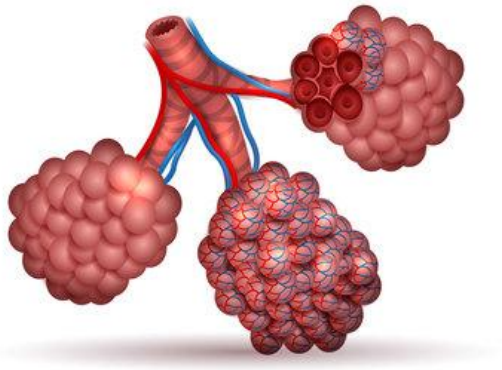
- At night, when there is no photosynthesis occurring, $CO_2$ elimination is the major exchange activity.
- During the day,$CO_2$ generated during respiration is used up for photosynthesis, hence oxygen release is the major event at this time.

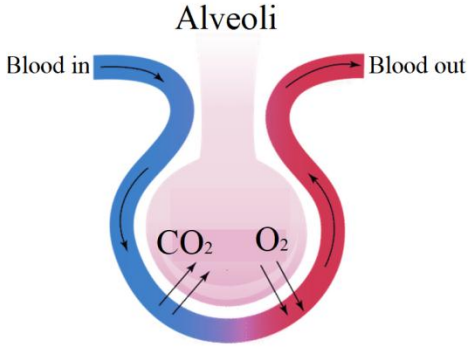
Terrestrial animals can breathe the oxygen in the atmosphere, but animals that live in water need to use the oxygen dissolved in water.

- The amount of dissolved oxygen is fairly low compared to the amount of oxygen in the air, hence the rate of breathing in aquatic organisms is much faster than that seen in terrestrial organisms.

## Human Respiratory system:

- In human beings, air is taken into the body through the nostrils.
- The air passing through the nostrils is filtered in the passage by fine hairs and mucus.
- From here, the air passes through the throat and into the lungs. Rings of cartilage are present in the throat. These ensure that the air-passage does not collapse.
- Within the lungs, the passage divides into smaller and smaller tubes which finally terminate in balloon-like structures called alveoli.
- The alveoli provide a surface where the exchange of gases can take place. The walls of the alveoli contain an extensive network of blood-vessels.






## Breathing cycle :

**Inspiration (Breath IN) :** Inspiration is the active intake of air from atmosphere into lungs. When we breathe in, we lift our ribs and flatten our diaphragm, and the chest cavity becomes larger as a result.

Pathway : Nostrils → Nasal cavity → Pharynx → Larynx → Trachea → Bronchi → Bronchioles → Alveoli

**Expiration (Breath OUT):** It is the passive expelling of air from the lungs. When the lungs exhale, the diaphragm relaxes, and the volume of the thoracic cavity decreases, As a result, the lungs contract and air is forced out.

Pathway : Alveoli → Bronchioles → Bronchi → Trachea → Larynx → Pharynx → Nasal cavity → Nostrils

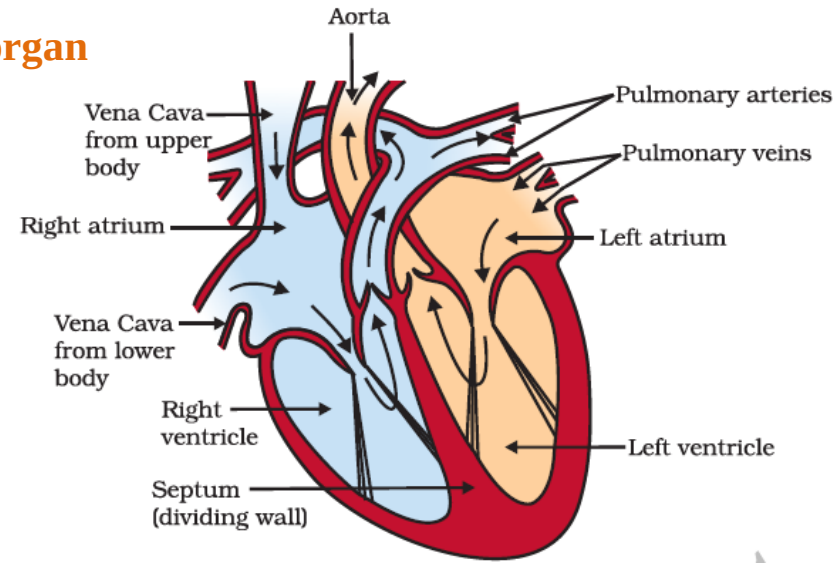
- In humans, the respiratory pigment haemoglobin present in red blood cells carries oxygen from lungs to different tissues of the body. Haemoglobin has more affinity towards oxygen.
- Carbon dioxide is more soluble in water than oxygen is and hence is mostly transported by the plasma.

# TOPIC-3 TRANSPORTATION

## Transportation in Human Beings

- Blood is a transporting media, it transports food, oxygen and waste materials in our bodies.
- Blood consists of a fluid medium called plasma in which the cells are suspended. Plasma transports food, carbon dioxide and nitrogenous wastes in dissolved form. Oxygen is carried by the red blood corpuscles.
- We need a pumping organ (heart) to push blood around the body, a network of tubes (blood vessels) to reach all the tissues.

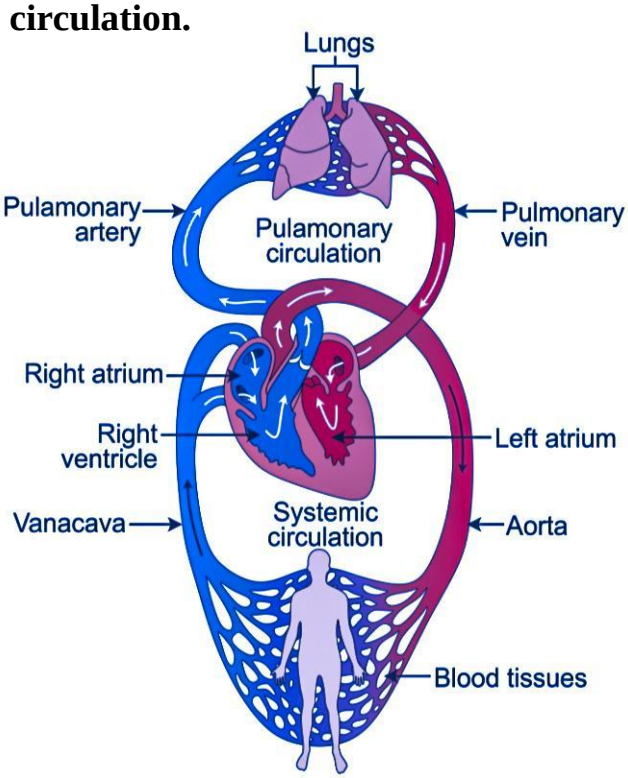
## HEART - the pumping organ




- The heart has four different chambers to prevent the oxygen-rich blood from mixing with the blood containing carbon dioxide.
- Oxygen-rich blood from the lungs comes to the thin-walled left atrium.
- The left atrium relaxes when it is collecting this blood. It then contracts, while the next chamber, the left ventricle, relaxes, so that the blood is transferred to it. When the muscular left ventricle contracts in its turn, the blood is pumped out to the body.
- De-oxygenated blood comes from the body to the thick walled right atrium, as it relaxes. As the right atrium contracts, the corresponding lower chamber, the right ventricle, dilates.
- This transfers blood to the right ventricle, which in turn pumps it to the lungs for oxygenation.
- Since ventricles have to pump blood into various organs, they have thicker muscular walls than the atria.
- Valves ensure that blood does not flow backwards when the atria or ventricles contract.

## Necessity of having four chambered heart :

- The heart has four different chambers to prevent the oxygen-rich blood from mixing with the blood containing carbon dioxide. Such separation allows a highly efficient supply of oxygen to the body. This is useful in animals that have high energy needs, such as birds and mammals, which constantly use energy to maintain their body temperature. Such animals are called **warm blooded animals.**
- In animals that do not use energy for this purpose, the body temperature depends on the temperature in the environment. Such animals are called **cold blooded animals ,** like amphibians or many reptiles have three-chambered hearts, and tolerate some mixing of the oxygenated and de-oxygenated blood.
- Fishes have only two chambers to their hearts, and the blood is pumped to the gills, is oxygenated there, and passes directly to the rest of the body. Thus, blood goes only once through the heart in the fish during one cycle.
- In other vertebrates, it goes through the heart twice during each cycle. This is known as **double circulation.**




## BLOOD VESSELS :

- **Arteries** are the vessels which carry blood away from the heart to various organs of the body. Since the blood emerges from the heart under high pressure, the arteries have thick, elastic walls.

- **Veins** collect the blood from different organs and bring it back to the heart. They do not need thick walls because the blood is no longer under pressure, instead they have valves that ensure that the blood flows only in one direction.
- The artery divides into very smaller vessels to bring the blood in contact with all the individual cells. The smallest vessels have walls which are one-cell thick and are called **capillaries**.
- Exchange of material between the blood and surrounding cells takes place across the thin wall of capillary. These capillaries then join together to form veins that convey the blood away from the organ or tissue.
- Leakage in the blood vessels lead to a loss of pressure, which would reduce the efficiency of heart. To avoid this, the blood has **platelet cells** which circulate around the body and plug these leaks by helping to clot the blood at the point of injury.

### Lymph

- There is another type of fluid also involved in transportation. It carries digested and absorbed fat from intestine and drains excess fluid from extra cellular space back into the blood.
- Through the pores present in the walls of capillaries some amount of plasma, proteins and blood cells escape into intercellular spaces in the tissues to form the tissue fluid or lymph.

## Transportation in Plants

Plants have low energy needs, and can use relatively slow transport systems compared to animals. Because,

- Plants do not move
- Plant bodies have a large proportion of dead cells in many tissues.

Plant transport system comprises, -

- Transportation of water and minerals
- Transportation of food (products of photosynthesis)

### Transportation of water

- It is carried out by the xylem tissue.
- Evaporation in leaf creates a suction pressure which pulls water from the xylem cells of roots.
- The loss of water in the form of vapour from the aerial parts of the plant is known as **transpiration.**

### Role of transpiration :

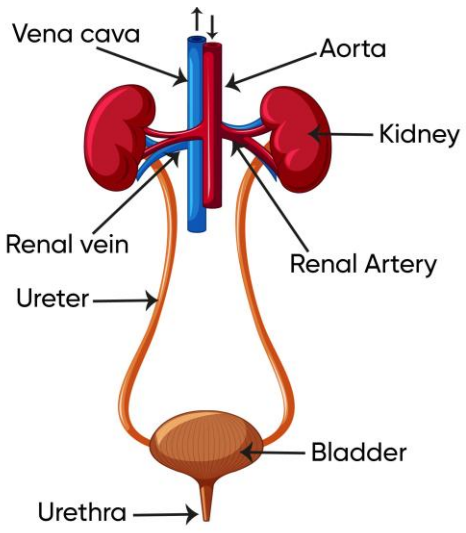
- Transpiration helps in the absorption and upward movement of water and minerals dissolved in it from roots to the leaves.
- It helps in temperature regulation.

## Transportation of food

- It is carried out by the phloem tissue.
- Products of photosynthesis, are moved from leaves to other parts of the plant.
- The transportation of soluble products of photosynthesis is called translocation
- These substances are especially delivered to the storage organs of roots, fruits and seeds and to growing organs.
- The translocation of food and other substances takes place in the sieve tubes with the help of adjacent companion cells both in upward and downward directions
- The translocation in phloem is achieved by utilizing energy.
- Increases in the osmotic pressure of the tissue causes the movement of food material.

# TOPIC-4 EXCRETION

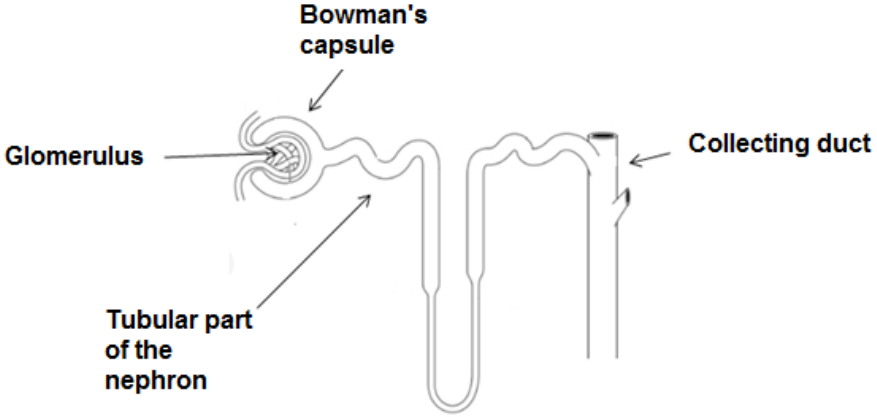
Excretion is a process, in which the harmful metabolic nitrogenous wastes like **urea** and **uric acid** generated are removed from the body.




Excretory system of human beings includes a pair of Kidney, a Urinary Bladder, a pair of Ureter and a Urethra.

Each kidney contains many filtration units called nephrons. Nephrons are the basic filtration units of kidneys. They carry out filtration, selective reabsorption and tubular secretion

### Structure Of the Nephron :



A nephron is made up of a cluster of thin walled capillaries called glomerulus which is associated with a cup like structure called as Bowman's capsule. The capsule has a long tube which terminates in to collecting duct.

**Urine formation involves three steps:**

**(i) Glomerular filtration:** Nitrogenous wastes, glucose, water, amino acids filter from the blood into Bowman's capsule of the nephron.

**(ii) Tubular reabsorption :** Useful substances from the filtrate are re-absorbed back by capillaries surrounding the nephron.

**(iii) Secretion :** Urea, extra water and salts are secreted in the tubule which open up into the collecting duct and then into the ureter.

Finally each ureter drains the urine in the urinary bladder where it is stored until the pressure of expanded bladder leads to an urge to pass it out through urethra.

The amount of water re-absorbed depends upon:

- How much excess of water is there in the body and,
- How much nitrogenous wastes need to be excreted out.

## Excretion in Plants

(i) The plants get rid of gaseous products-through stomata in leaves and lenticels in stems.

(ii) The plants get rid of stored solid and liquid waste by the shedding off leaves, peeling off bark and felling off fruits.

(iii) The plants get rid of wastes by secreting them in the form of gums and resins.

(iv) Plants also excrete some waste substances into the soil around them.

CHAPTER-6

# CONTROL AND COORDINATION

## TOPIC-1 Control and Coordination in Animals

### Need for a system of control and coordination in Animals

An organism needs control and coordination system -
(i) To save the body of the organisms from the harmful changes in the environment.
(ii) To control the speed of voluntary and involuntary actions.
(iii) To have the capability to think and learn for responding to any stimuli.

Control and coordination is brought about in all animals with the help of two main systems: **Nervous system** and **Endocrine system.**

## Nervous system

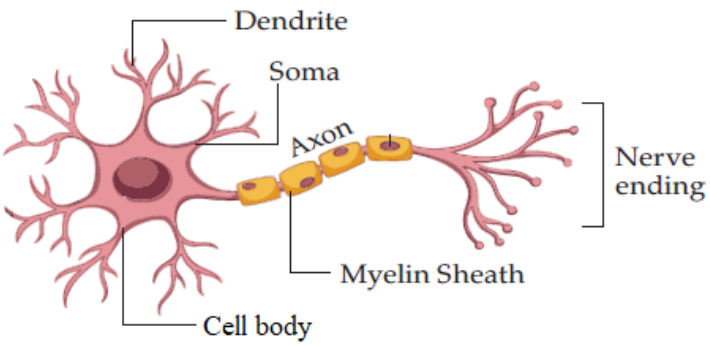
It is the system of conducting tissues that receives the stimulus and transmits it to other parts of the body forming a network of nerves. It is involved in receiving information and generating responses to that information.

All information from our environment is detected by the specialised tips of some nerve cells. These receptors are usually located in our sense organs, such as the ear, eye, nose, skin and the tongue.

- Gustatory receptors will detect taste
- Olfactory receptors will detect smell
- Phono receptors will detect sound
- Photo receptors will detect light
- Thermo receptors will detect heat / cold

The structural and functional units which make up the nervous system are called **nerve cells** or **neurons**.

### Structure and function of Neuron :



**(i) Cell body:** It is star shaped which contains nucleus with abundant cytoplasm. The information acquired by it travels as an electrical impulse.

**(ii) Dendrite:** The hair like structure protruding out from margins of cell body is called dendrite. It receives the nerve impulses.

**(iii) Axon:** It is the longest fiber on the cell body. It ends in several hairs like structures called axon terminals, which transmits electrical impulse from cell body to dendrite of next neuron.

**(iv) Myelin sheath:** It is an insulator covered around the axon.

**(v) Synapse:** It is the point of contact between the nerve ending of one neuron and dendrite of other neuron. It is the part where electrical signal is converted into chemical signal

**Nerve Impulse:** It is the information in the form of chemical and electrical signals passing through neurons. These impulses are carried by dendrites towards the cell body.

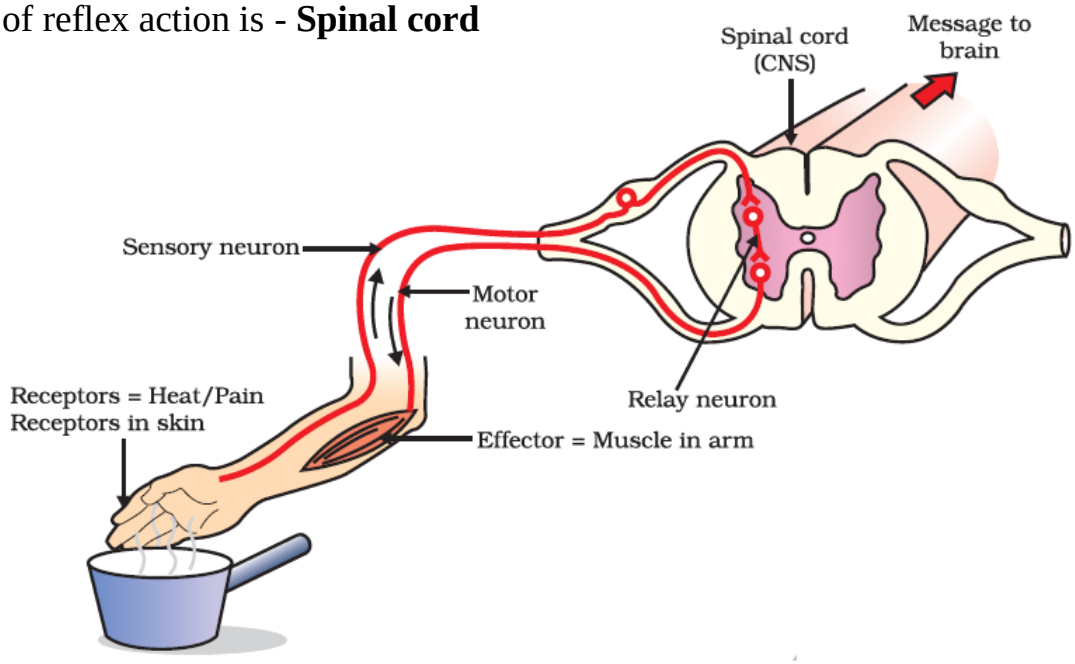
**The pathway of nerve impulse :**

Dendrites → Cell body → Axon → Nerve endings → Synapse → Dendrite of next neuron

**Reflex action:** It is quick, sudden and immediate response of the body to a stimulus. e.g., withdrawal of hand on touching hot object.

**Reflex arc:** The pathway through which nerve impulse passes during reflex action is called reflex arc i.e., it is a pathway through which the reflex action occurs.

The site of reflex action is - **Spinal cord**



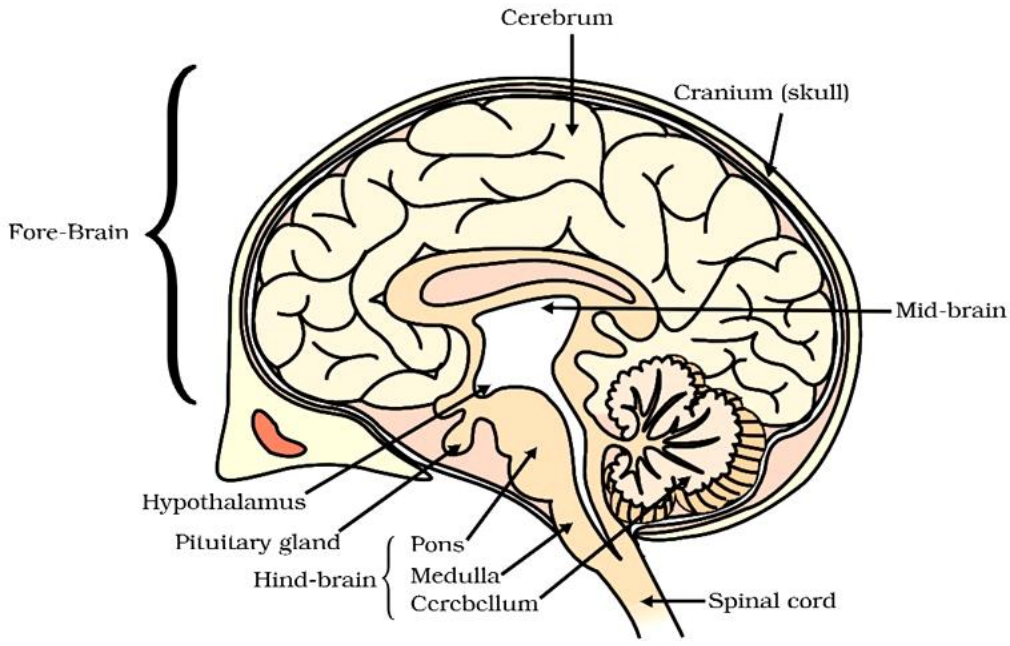
## Responses are of three main types :

**(a) Voluntary:** Controlled by forebrain. For e.g., talking, writing, walking, etc.
**(b) Involuntary:** Controlled by midbrain and hind-brain. For e.g., heartbeat, vomiting, respiration, etc.
**(c) Reflex action:** Controlled by spinal cord. For e.g., withdrawal of hand on touching a hot object.

## Need of Reflex Actions:

In some situations such as touching a hot object, pinching, etc., we need to act quickly, otherwise our body would be harmed. Hence, this response is generated from spinal cord instead of brain.

# Human nervous system




Human brain is the main coordinating centre of the body. The brain is protected by the skull called the cranium.
It has three major parts : Forebrain, midbrain and hind brain.

**(a) Fore-brain:** It is the most complex or specialized part of the brain. It consists of **cerebrum.** The main functions of forebrain are as follows:

(i) Main thinking part of the brain.
(ii) Controls the voluntary actions.
(iii) Stores information (memory).
(iv) Receives sensory impulses from various parts of the body and integrate it.
(v) It is the centre associated with hunger.

**(b) Mid-brain:** Controls involuntary actions.

**(c) Hind-brain:** It has three parts:

**(i) Cerebellum:** Controls posture and balance, precision of voluntary actions. e.g., picking pen.
**(ii) Medulla:** Controls involuntary actions. e.g., blood pressure, salivation, vomiting.
**(iii) Pons:** Controls voluntary actions and helps in regulation of respiration.

# Endocrine system

Endocrine system comprises endocrine glands and its secretions, called hormones.

Hormones are chemical messengers secreted in very small amounts by specialized tissue called Endocrine (ductless) glands. They act on target tissues/organs usually away from their source.

## Endocrine glands : Their secretions and function

### Pituitary gland :

- It secretes Growth hormone.
- Growth hormone regulates growth and development of the body.
- If there is a deficiency of this hormone in childhood, it leads to dwarfism(very short ) and over secretion of this hormone leads to Gaigantism(extremely tall).

### Thyroid gland :

- It secretes thyroxin hormone.
- Thyroxin regulates carbohydrate, protein and fat metabolism in the body so as to provide the best balance for growth.
- Iodine is essential for the synthesis of thyroxin. In case iodine is deficient in our diet, there is a possibility that we might suffer from goitre. One of the symptoms in this disease is a swollen neck.
- So, it is important for us to have iodised salt in our diet

### Adrenal gland :

- It secretes adrenalin hormone.
- This hormone is also called as ‘fight’ or ‘flight’ hormone or ‘Emergency’ hormone. Because, it prepares our body for scary situations.
- Adrenaline is secreted directly into the blood and carried to different parts of the body. As a result, the heart beats and rate of breathing will be faster, resulting in supply of more oxygen to our muscles.
- All these responses together enable our body to be ready to deal with the situation.

### Pancreas :

- Pancreas gland is dual in nature as it is both exocrine as well as endocrine gland. Hence, it is called as mixed gland.
- It secretes insulin hormone.
- This hormone regulates the blood sugar level.
- If it is not secreted in proper amounts, the sugar level in the blood rises causing diabetes.
- To treat increased level of blood sugar, the diabetic patients are treated by giving injections of insulin.

### Sex hormones :

- **Testosterone** in males and **Oestrogen** in females brings changes associated with puberty.
- Testosterone is secreted by testis and it promotes appearance of male sex characters like voice deepening and the growth of armpit, chest and pubic hair.
- Testosterone promotes the production of sperms.
- Oestrogen is secreted by ovaries and it promotes appearance of female sex characters like Enlargement of breasts and erection of nipples. Growth of body hair, most prominently underarm and pubic hair. Widening of hips
- Oestrogen promotes the production of eggs and regulates the menstrual cycle.

### Feedback mechanism:

The excess or deficiency of hormones have a harmful effect on our body. The timing and amount of hormone released are regulated by feedback mechanisms.

For example, if the sugar levels in blood rise, they are detected by the cells of the pancreas which respond by producing more insulin. As the blood sugar level falls, insulin secretion is reduced.

## TOPIC-2 Control and Coordination in Plants

Plants respond and react to the changes that happen in the environment around them.

The changes in the environment to which the plant respond and react are called stimuli such as light, gravity, chemicals, touch, etc.

**Tropic Movement**: It is the directional growth movement of a plant organ in response to an external stimulus. Growth towards the stimulus is positive tropism and growth away from the stimulus is negative tropism.

Plants show two different types of movement :

**(A) Growth independent movements:** These are directionless movements of plants caused due to variations in the outside environment. e.g., Folding or drooping of leaves of "Touch me not plant" on touching it. This is known as thigmotropism.

Some cells must change shape in order for movement to happen. Cells change their shape by changing the amount of water in them, resulting in swelling or shrinking

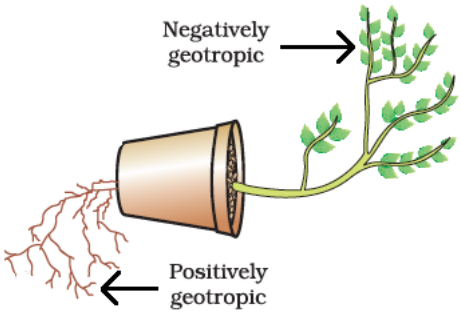
**(B) Growth-dependent movements:** These are directional movements of plants caused due to the presence of a specific stimulus.

It is of four types:

**(i) Phototropism:** Growth movements of plants towards light e.g., shoots bend toward light (positively phototropic) and roots move away from light (negatively phototropic).

**(ii) Geotropism:** Movement towards gravity.

e.g., Roots of a plant are positively geotropic while shoots of a plant are negatively geotropic.




**(iii) Chemotropism:** Movement towards chemicals. e.g., Growth of pollen tube towards ovule.

**(iv) Hydrotropism:** Movement towards water. e.g., Movement of plant roots towards water. It is positively hydrotropism.

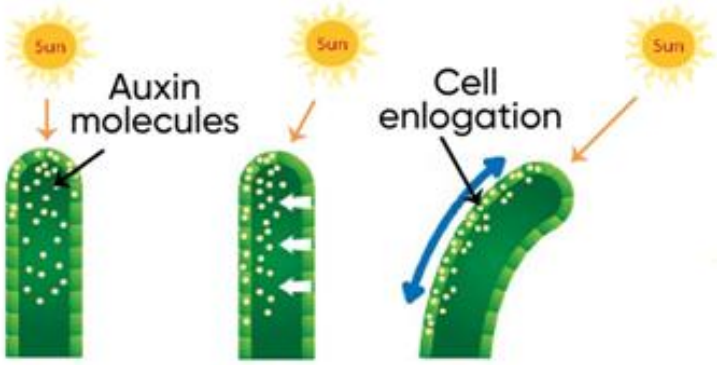
## Plant hormones

Hormones which helps to coordinate growth ,development and response to the environment are called plant hormones or phtohormones

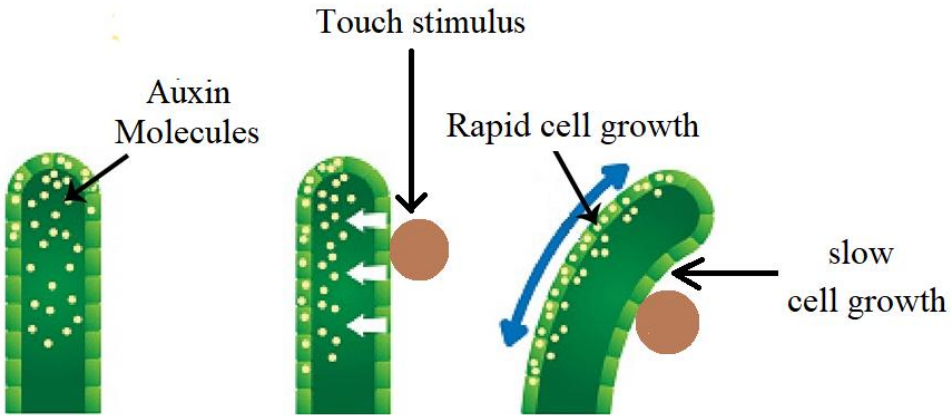
| | |
|---|---|
| Auxins | Synthesized at shoot tip.<br>Helps the cells to grow longer.<br>Involved in tropic movements of plants. |
| Gibberellin | Helps in the growth of the stem. |
| Cytokinins | Promotes cell division.<br>Present in greater concentration in fruits and seeds. |
| Abscisic Acid | Inhibits growth.<br>Causes wilting of leaves. |

## Action of Auxins :

- When growing plants detect light, a hormone called auxin, synthesized at the shoot tip, helps the cells to grow longer.
- When light is coming from one side of the plant, auxin diffuses towards the shady side of the shoot.
- This concentration of auxin stimulates the cells to grow longer on the side of the shoot which is away from light.
- Thus, the plant appears to bend towards light.




## Role of Auxin in growth of tendrils :




When a tendril comes in contact with a support, auxin stimulates faster growth of the cells on the opposite side that's why the tendril forms a coil around the support.

CHAPTER-7

# HOW DO ORGANISMS REPRODUCE

## TOPIC-1 ASEXUAL REPRODUCTION

- Reproduction is the process by which living organisms produce new individuals similar to themselves. It ensures continuity of life on the earth.
- Chromosomes in the nucleus of a cell contain information for inheritance of features from parents to next generation in the form of DNA.
- Basic event in reproduction is the creation of a DNA copy.
- The process of copying the DNA will have some variations each time. As a result, the DNA copies generated will be similar, but may not be identical to the original.
- This inbuilt tendency for variation during reproduction is the basis for evolution

### The Importance of Variation

- All organisms are living in a particular niche (habitat)
- If the niche were drastically altered, the population of an organism could be wiped out.
- So, variations in few individuals in a population, there would be some chance for them to survive.
- Variation is thus useful for the survival of species over time.

**Example :** Bacteria living in temperate waters, and if the water temperature were to be increased by global warming, most of these bacteria would die, but the few variants resistant to heat would survive and grow further.

### Types of Reproduction

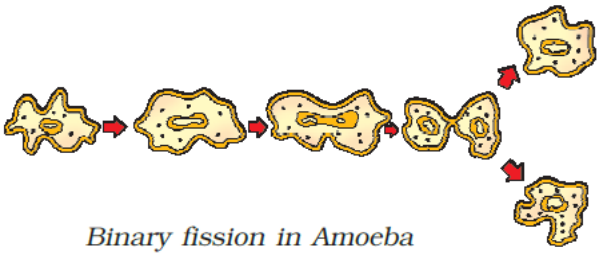
| Asexual reproduction | Sexual reproduction |
|---|---|
| A single individual give rise to new individual. | Two individuals i.e., one male and one female are needed to give rise to new individual. |
| Gametes are not formed. | Gametes are formed. |
| New individuals are identical to their parents. | New individual is genetically similar but not identical to parents. |
| Asexual reproduction is seen in most of the lower organisms and plants as well. | Adopted by higher organisms. |

## Modes of Asexual reproduction

**Fission :** It is of two types - binary fission and multiple fission.

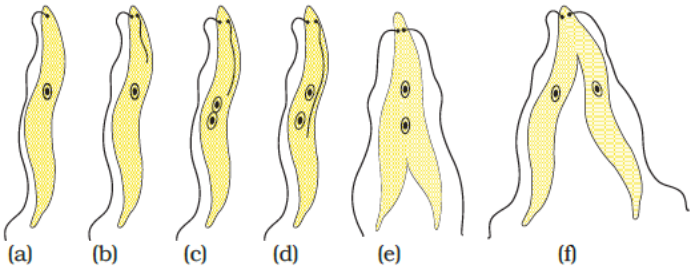
**(i) Binary fission:** It is the division of one cell into two similar or identical cells. The nucleus first divides automatically into two, followed by the division of the cytoplasm. The cell finally splits into two daughter cells. e.g., Amoeba , Leishmania

In organisms such as Amoeba, the splitting of the two cells during division can take place in any plane.



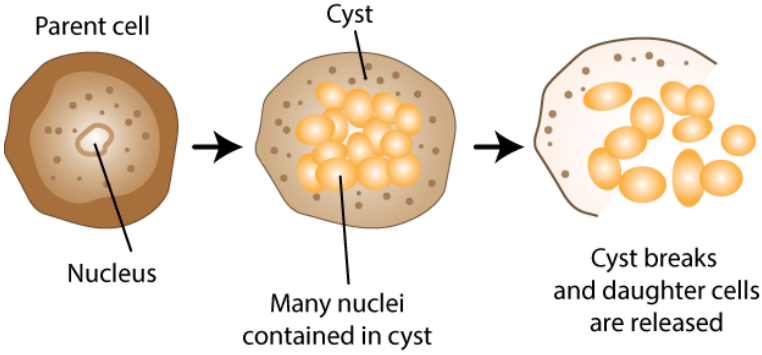
*Binary fission in Amoeba*

However, some unicellular organisms show somewhat more organisation of their bodies, such as is seen in Leishmania, which have a whip-like structure at one end of the cell. In such organisms, binary fission occurs in a definite orientation

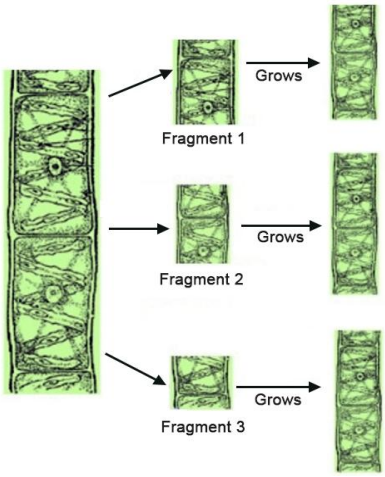



*Binary fission in Leishmania*

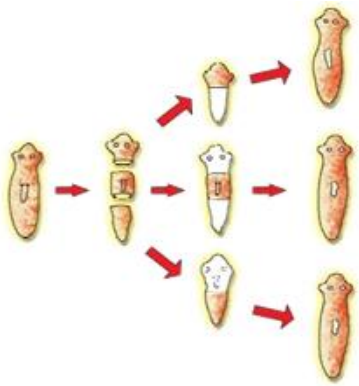
**(ii) Multiple fission:** In multiple fission, many individuals are formed from a single individual. e.g., Plasmodium. The nucleus divides repeatedly, producing many nuclei and many daughter cells are formed.



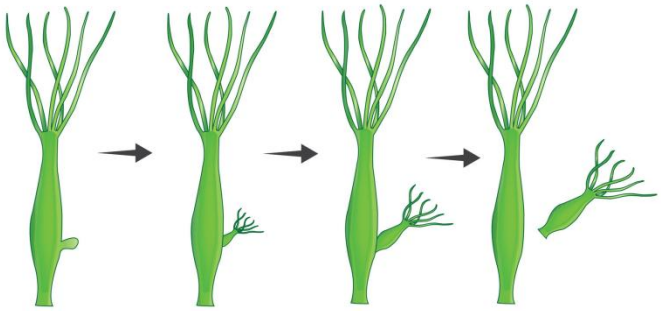

**Fragmentation:** It takes place in multi cellular organisms with simple body organisation such as in Spirogyra. In this, the body breaks up into two or more small pieces of fragments upon maturation. These fragments grow into new individuals.



**Regeneration:** It is the ability of a fully differentiated organism to give rise to new individual organisms from its body parts. Small cut or broken parts of the organism's body grow or regenerate into separate individuals. For example: Planaria and Hydra.



**Budding:** In budding, a small part of the body of the parent grows out as a bud which then detaches and becomes a new organism. Hydra reproduces by budding using the regenerative cells.



Budding in Hydra

**Spore Formation:** Spores are small bulb like structures which are covered by thick walls. Under favourable conditions, they germinate and produce new organisms. e.g., Rhizopus.

**Vegetative Propagation:** In many plants, new plantlets develops from vegetative parts of a plant's body such as stem, roots, leaves, etc.

### Advantages of Vegetative propagation :

- Seedless plants can be grown through this method.
- Plants produced by layering and grafting can bear flowers and fruits earlier than those produced from seeds.
- Plants produced from this method are genetically similar
- It is cheaper, easier and more rapid method of plant propagation.

## SEXUAL REPRODUCTION :

- The sexual mode of reproduction incorporates a process of combining DNA from two different individuals during reproduction. This brings more variation in an individual.
- To attain genetic stability, sexually reproducing organisms should produce germ cells (gamete) having only half the number of chromosomes and half the amount of DNA as compared to the non-reproductive body cells.

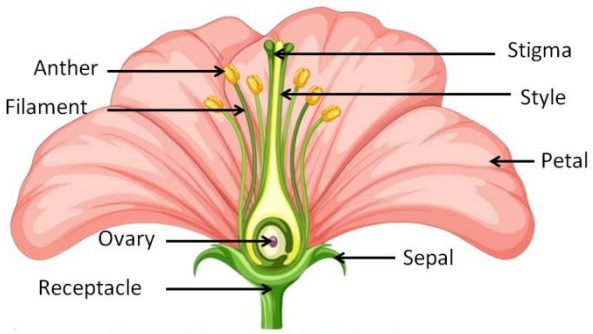
- This is achieved by a process of cell division called meiosis. Thus, when these germ-cells from two individuals combine during sexual reproduction to form a new individual, it results in reestablishment of the number of chromosomes and the DNA content in the new generation.

**Nature of gametes :**

- If the zygote is to grow and develop into an organism which has highly specialised tissues and organs, then it has to have sufficient stores of energy for doing this.
- One germ-cell is large and contains the food-stores while the other is smaller and likely to be motile.
- Conventionally, the motile germ cell is called the male gamete and the germ-cell containing the stored food is called the female gamete.

## TOPIC-2 REPRODUCTION IN FLOWERING PLANTS

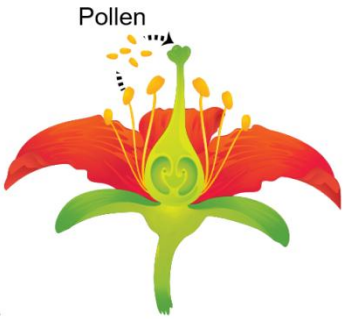
- The reproductive parts of angiosperms are located in the flower.
- Stamens and pistil are the reproductive parts of a flower which contain the germ-cells.



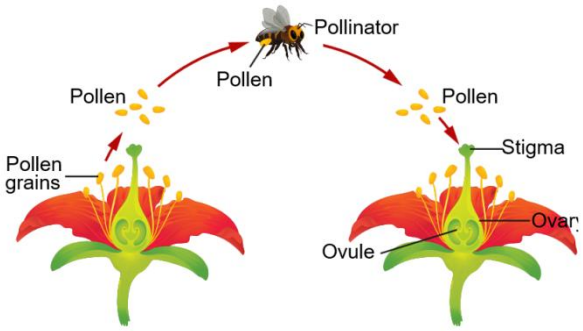
PARTS OF A FLOWER

- The flower may be unisexual when it contains either stamens or pistil Ex: Papaya, watermelon
- The flower may be bisexual when it contains both stamens and pistil. Ex : Hibiscus, mustard
- Stamen is the male reproductive part and it produces pollen grains that are yellowish in colour.
- Pistil is present in the centre of a flower and is the female reproductive part. It is made of three parts.
- The swollen bottom part is the ovary, middle elongated part is the style and the terminal part which may be sticky is the stigma.
- The male germ-cell produced by pollen grain fuses with the female gamete present in the ovule. This fusion of the germ-cells or fertilisation gives us the zygote which is capable of growing into a new plant.
- The pollen needs to be transferred from the stamen to the stigma.

- If this transfer of pollen occurs in the same flower, it is referred to as **self-pollination**.
- If the pollen is transferred from one flower to another, it is known as **crosspollination.**

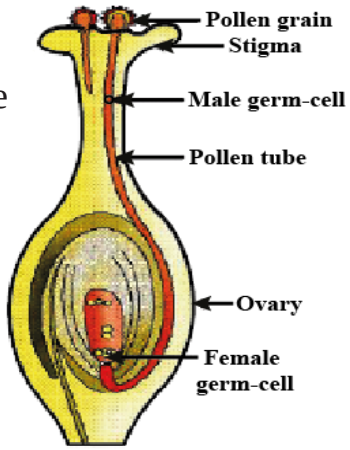



SELF POLLINATION

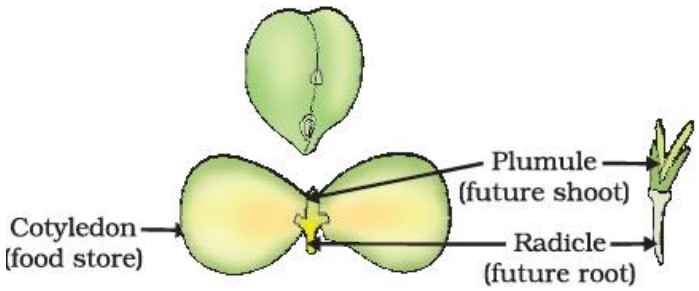



CROSS POLLINATION

- The transfer of pollen from one flower to another is achieved by agents like wind, water or animals.
- After the pollen lands on a suitable stigma, it has to reach the female germ-cells which are in the ovary. For this, a tube grows out of the pollen grain and travels through the style to reach the ovary.




- After fertilisation, the zygote divides several times to form an embryo within the ovule.
- The ovule develops a tough coat and is gradually converted into a seed.
- The ovary grows rapidly and ripens to form a fruit.
- Meanwhile, the petals, sepals, stamens, style and stigma may shrivel and fall off.
- The seed contains the future plant or embryo which develops into a seedling under appropriate conditions. This process is known as germination.



# TOPIC-3 REPRODUCTION IN HUMAN BEINGS

It needs sexual maturation, which is the period of life when production of germ cells i.e., ova (female) and sperm (male) start in the body. This period of sexual maturity is called **puberty**.

In early teenage years, a whole new set of changes occurs.

**Some of these changes are common to both boys and girls.**

- Thick hair growth in armpits and genital area.
- Skin becomes oily, may result in pimples.

**In girls:**

- Breast size begins to increase.
- Girls begin to menstruate.

**In boys:**

- Thick hair grows on face in the form of beard and moustache.
- Voice begins to crack.
- Production of sperms

## Male Reproductive System

### (a) Testes :

- A pair of testes are located inside scrotum which is present outside the abdominal cavity, because sperm formation requires a lower temperature than the normal body temperature.
- The formation of germ-cells or sperms takes place in the testes.
- Testes release male sex hormone - testosterone, which regulates the production of sperms.

### (b) Vas deferens :

- The sperms formed are delivered through the vas deferens which unites with urethra which is a common passage for both the sperms and urine.

### (c) Associated glands:

- Seminal vesicles and prostate gland are associated glands, which add their secretion to the sperms. This fluid provide nourishment to sperms and make their transport easy.

# Female Reproductive System

## (a) Ovary:

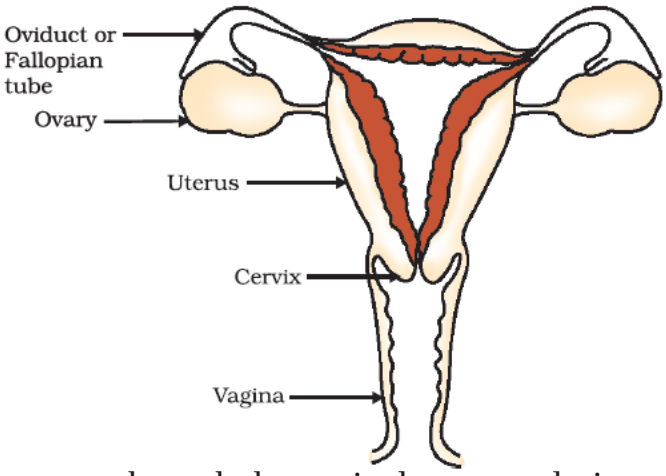
- A pair of ovary is located in both sides of abdomen.
- Female germ cells i.e., eggs are produced here.
- Ovaries release female sex hormone-Oestrogen which regulates the production of eggs and menstrual cycle.

## (b) Oviduct or Fallopian tube

- Receives the egg produced by the ovary and transfer it to the uterus.
- Fertilisation i.e., fusion of gametes takes place here.

## (c) Uterus:

- It is a bag-like structure where development of the foetus takes place.




- The sperms enter through the vaginal passage during sexual intercourse. They travel upwards and reach the oviduct where they may encounter the egg.
- The fertilised egg (zygote) starts dividing and form a ball of cells or embryo.
- The embryo is implanted in the lining of the uterus where they continue to grow and develop organs to become foetus.

## Placenta :

- The embryo gets attached to the uterus with the help of a special tissue called placenta. This is a disc which is embedded in the uterine wall. It contains villi on the embryo's side of the tissue.
- It provides nourishment to the foetus.
- The waste generated by developing embryo is removed by this structure.

The development of the child inside the mother's body takes approximately nine months. The child is born as a result of rhythmic contractions of the muscles in the uterus.

# Menstruation:

- The ovary releases one egg every month, the uterus also prepares itself every month
- to receive a fertilised egg.
- Thus its lining becomes thick and spongy. This would be required for nourishing the embryo if fertilisation had taken place.
- If the egg is not fertilised, this lining is not needed any longer. So, the lining slowly breaks and comes out through the vagina as blood and mucous.
- This cycle takes place roughly every month and is known as menstruation.
- It usually lasts for about two to eight days.

## Sexually Transmitted Diseases (STDs) :

These are the diseases which spread by sexual contact from an infected person to a healthy person.

Some common STDs are -

- **Bacterial :** Gonorrhoea, syphilis,
- **Viral :** warts, HIV-AIDS.

## Birth Control Methods:

There are different methods which are developed to prevent and control pregnancy such as mechanical methods, chemical methods, oral pills and surgical methods. These are also known as methods of contraception.

### (a) Physical barrier :

- To prevent union of egg and sperm.
- Use of condoms, cervical caps and diaphragm.

### (b) Chemical methods :

- Use of oral pills.
- These change hormonal balance of body so that eggs are not released.
- May have side effects.

### (c) Intrauterine contraceptive device (IUCD)

- Copper-T or loop is placed in uterus to prevent pregnancy.

### (d) Surgical methods

- In males the vas deferens is blocked to prevent sperm transfer and the process is called **vasectomy.**
- In females, the fallopian tube is blocked to prevent egg transfer and the process is called **tubectomy.**

CHAPTER-8

# HEREDITY

## Variation

- Variations arise usually during the process of reproduction.
- Organisms those reproduce asexually have few variations.
- Organisms those reproduce sexually have more variations.

Variation promotes the survival of species in the following ways:

i. Variation improves the survival rate of species
ii. It helps the individual organisms to adapt accordingly with the environmental conditions.

## Inherited Traits

The traits that are inherited from the parents to the offspring are called inherited traits. Ex : Hair, skin, eye colour,

- A child bears all the basic features of a human being. However, it does not look exactly like its parents, and human populations show a great deal of variation.
- Both the father and the mother contribute practically equal amounts of genetic material to the child. This means that each trait can be influenced by both paternal and maternal DNA.
- Thus, for each trait there will be two versions in each child.

  Ex : Free ear lobes and attached ear lobes.
  Straight hair and curly hair

## Mendel's Contributions

Mendel used a number of contrasting visible characters of garden peas –

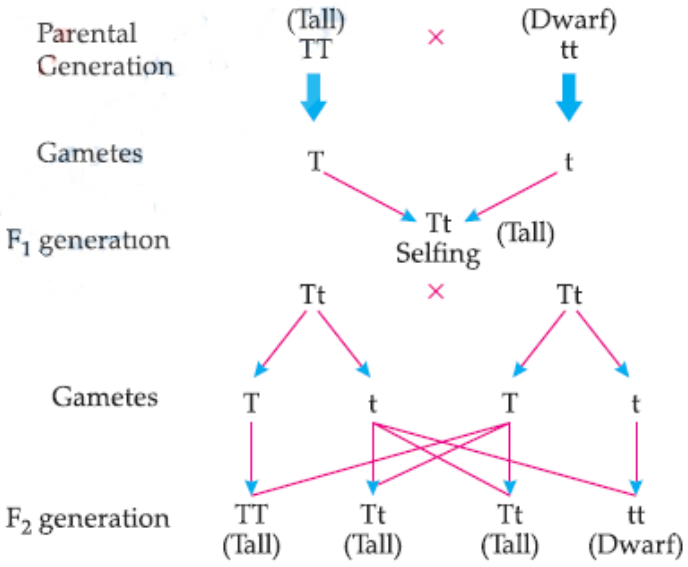
| Character | Dominant Trait | Recessive Trait |
|---|---|---|
| Flower colour | Violet | White |
| Flower position | Axial | Terminal |
| Seed colour | Yellow | Green |
| Seed shape | Round | Wrinkled |
| Pod shape | Inflated | Constricted |
| Pod colour | Green | Yellow |
| Height of plant | Tall | Dwarf/Short |

Mendel conducted a series of experiments in which he crossed the pollinated plants to study one character (at a time). Mendel conducted a series of experiments in which he crossed the pollinated plants to study one character (at a time).

## Monohybrid cross

Monohybrid cross is a cross between two pea plants with one pair of contrasting characters. e.g., Cross between a tall and a dwarf plant (short).

- He took pea plants with different characteristics – a tall plant and a short plant, produced progeny by crossing them,
- All plants were tall in first generation or $F_1$ progeny
- This meant that only one of the parental traits was seen, not some mixture of the two.
- Then plants of $F_1$ progeny were reproduced by self pollination.
- Second generation or $F_2$ progeny plants not all tall. Instead, one quarter of them are short.
- This indicates that both the tallness and shortness traits were inherited in the F1 plants, but only the tallness trait was expressed.



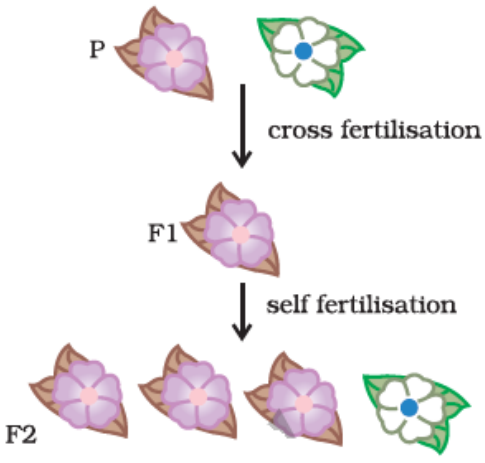

**Checker board :**

| | **T** | **t** |
|---|---|---|
| **T** | **TT** (Tall) | **Tt** (Tall) |
| **t** | **Tt** (Tall) | **tt** (Dwarf) |

Genotypic ratio : **1 : 2 : 1** Phenotypic ratio is : **3 : 1**

Both TT and Tt are tall plants, while only tt is a short plant. In other words, a single copy of 'T' is enough to make the plant tall, while both copies have to be 't' for the plant to be short. Traits like 'T' are called dominant traits, while those that behave like 't' are called recessive traits.

**Exercise :** When pea plant bearing pink flower (PP) is crossed with a pea plant plant bearing white flower, represent the result obtained in F2 generation of monohybrid cross with the help of checker board and mention the ratio of varieties of plants.




**Checker board :**

| | **P** | **p** |
|---|---|---|
| **P** | **PP** pink | **Pp** pink |
| **p** | **Pp** pink | **pp** white |

Genotypic ratio : **1 : 2 : 1** Phenotypic ratio is : **3 : 1**

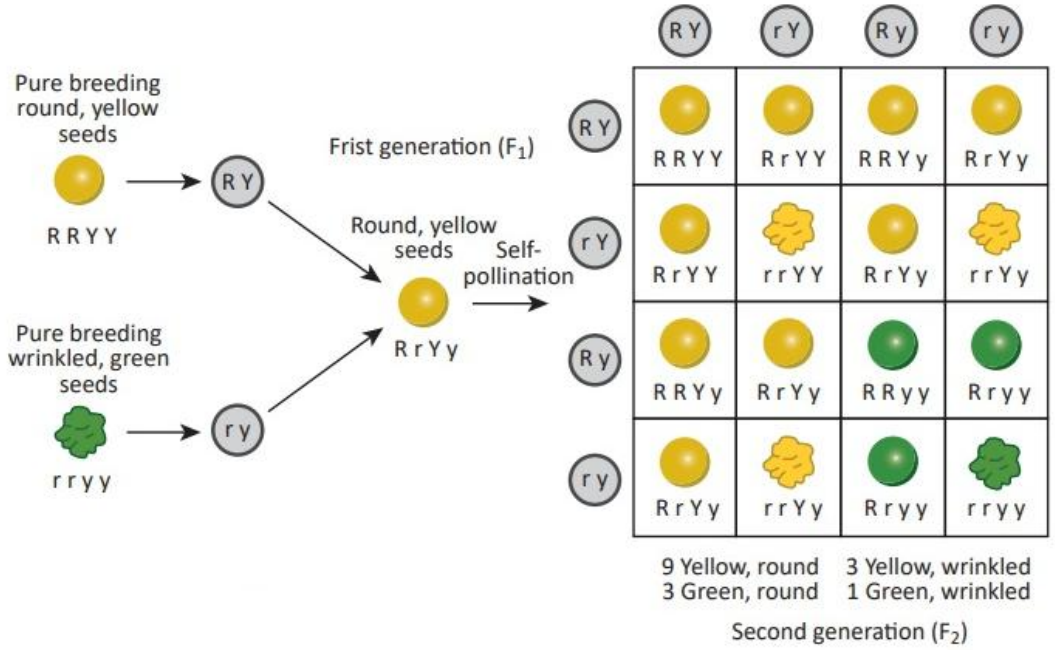
## Dihybrid cross

- Dihybrid cross is a cross between two pea plants with two pair of contrasting characters. e.g., Cross between plant bearing round yellow color seeds with plant bearing wrinkled green color seeds
- They are all have yellow and round seeds. Yellow color and round seeds are thus dominant traits.
- When these F1 progeny are used to generate F2 progeny by self-pollination, we will find the plants with the following combinations.

i. Yellow and round seeds
ii. Yellow and wrinkled seeds
iii. Green and round seeds
iv. Green and wrinkled seeds

Phenotypic ratio is : **9: 3 : 3 : 1**

From this result Mendel conclude that the yellow/green trait and the round seed/wrinkled seed trait are independently inherited.




## Expression of traits :

- Cellular DNA is the information source for making proteins in the cell.
- A section of DNA that provides information for one protein is called the gene for that protein.

Consider the example of tallness as a characteristic.

- Plant height could be depend on the amount of a particular plant hormone.
- The dominant gene triggers the secretion of hormone and plant will be tall. The recessive gene suppress the secretion of that hormone and plant will be short.

This means that each pea plant must have two sets of all genes, one inherited from each parent.

## How do germ-cells make a single set of genes?

- Genes are located on the chromosome. Thus, each cell will have two copies of each chromosome, one each from the male and female parents.
- Every germ cell will take one chromosome from each pair and these may be of either maternal or paternal origin.
- When two germ cells combine, they will restore the normal number of chromosomes in the progeny, ensuring the stability of the DNA of the species.

# Sex Determination

The process by which sex of a new born individual is determined is called **sex determination.**

Different species use very different strategies for sex determination.

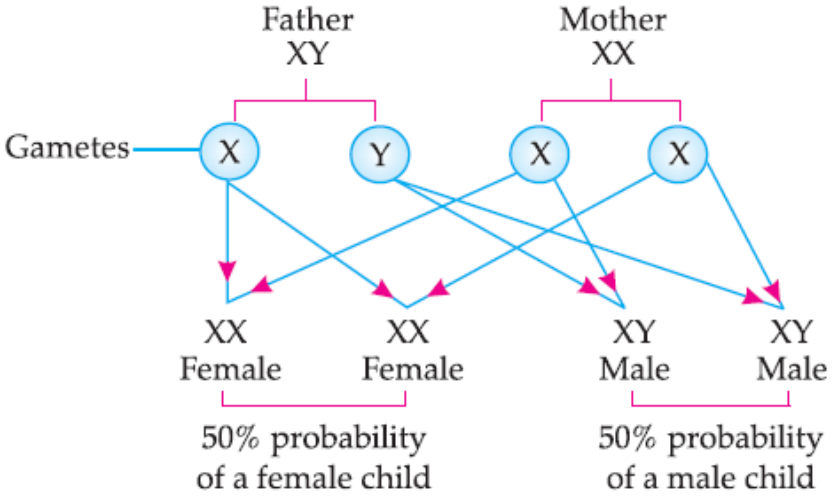
- In few reptiles temperature will determine whether the animals developing in the eggs will be male or female.

- In snails, individuals can change sex, indicating that sex is not genetically determined.
- In human beings, the sex of the individual is largely genetically determined.

## Sex determination in human beings:

- In human beings, there are 23 pairs of chromosomes. Out of these, 22 chromosome pairs are called autosomes and the last pair of chromosome which helps in deciding sex of the individual is called sex chromosome.
- Autosomes are pairs of chromosomes that are identical in appearance and are not involved in sex determination.
- Sex chromosomes are pairs of chromosomes involved in sex determination and are not identical in appearance. X is normal in size while Y is shorter.
- A male has one X and Y sex chromosomes (XY) while a female has two X-sex chromosomes (XX).
- Egg produced by female has X chromosome, Sperm produced by male have either X or Y chromosome.
- If an egg is fertilised by sperm carrying X-chromosome result into girl child.
- If an egg is fertilised by sperm carrying Y-chromosome result into girl child.
- Thus, the sex of the children will be determined by what they inherit from their father.

### Schematic representation :



CHAPTER-13

# OUR ENVIRONMENT

## TOPIC-1 ECOSYSTEM AND FOOD CHAIN

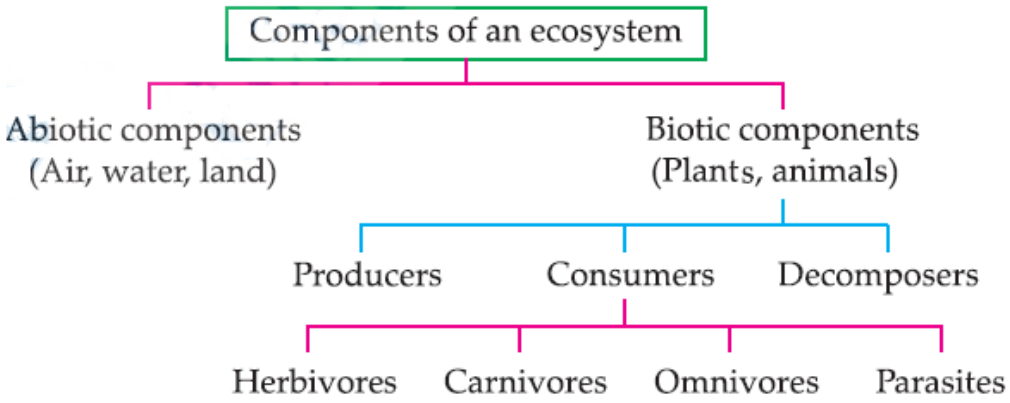
Everything that surrounds us is environment. It includes both living (biotic) and non-living (abiotic) components.

Interaction between these biotic and abiotic components forms an ecosystem.

**Types of Ecosystem :** It is of two types:

**(a) Natural Ecosystem:** The ecosystem which exists in nature on its own. e.g., forest, lake, ocean, etc.

**(b) Artificial Ecosystem:** Man-made ecosystem is called *artificial ecosystem*. e.g., crop field, aquarium, garden, etc.




**Producers** includes green plants and algae. They contain chlorophyll pigment which helps them to carry out the process of photosynthesis in the presence of light. Thus, they are also called as converters or transducers.

**Consumers** are those organisms which depend upon the producers for food, either directly or indirectly by feeding on other consumers for their sustenance. They are also called heterotrophs. Herbivores, carnivores, omnivores and parasites are the various types of consumers.

**Decomposers** are those micro-organisms that obtain energy from the chemical breakdown of dead organisms or animals or plant wastes. Decomposers break down the complex organic substances into simple inorganic substances that go into the soil and are used up again by the plants. They help in the replenishment of natural resources.

## Food chain

Food chain is the sequence of organisms through which food energy flows in an ecosystem. It is a succession of organisms that eat other organisms and may, in turn, be eaten themselves.

Example:

Grass → Grasshopper → Frog → Snake → Eagle
(Producer) (Herbivore) (Carnivore) (Carnivore) (Top Carnivore)



### Forest ecosystem :

Grass → deer → Tiger

### Pond ecosystem :

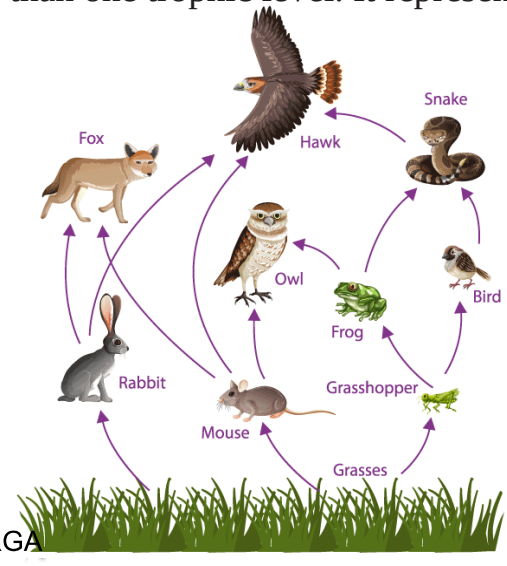
Phytoplankton → zooplankton → Larva → small fish → large fish

### Trophic Levels

Trophic Levels are the various steps or levels in the food chain where transfer of food or energy takes place. Producers are the first trophic level, herbivores are the second trophic level, carnivores or secondary consumers are the third trophic level and large carnivores or tertiary consumers are the fourth trophic level.

## Food Web

Food Web is the network of various food chains which are interconnected at various trophic levels. Since an organism can occupy position in more than one food chain, in a food web it occupies more than one trophic level. It represents the feeding relationship within the community.



## Energy Flow :

The flow of energy through different steps in the food chain is unidirectional. This means that the energy that is captured by the autotrophs does not revert back and the energy which passes to the herbivores does not come back to autotrophs.

At every step in a food chain the energy received by the organism is also used for its own metabolism and maintenance. The left over is passed to next higher trophic level. Thus, energy flow decreases with successive trophic levels.

The number of steps is limited to four or five in a food chain for the transfer of energy. Because, very less amount of energy is available to the last trophic level.

## 10 Percent Law:

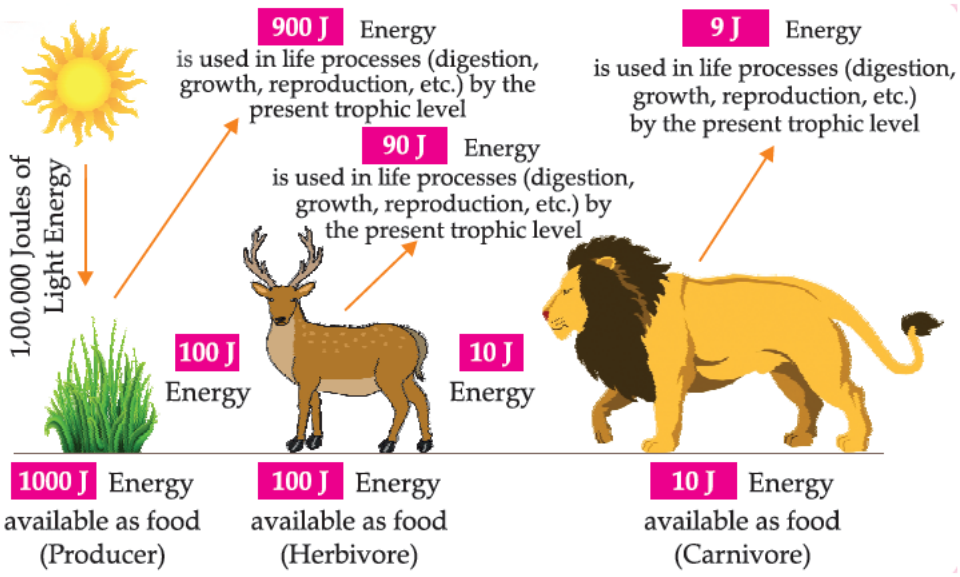
It states that only 10 per cent of food energy is transferred from one trophic level to the next level. The remaining 90 per cent energy is used in life processes by the present trophic level.

Due to this gradual decrease in energy, food chains contain 3 - 4 trophic levels.

## Biological Magnification :

The concentration of harmful chemicals goes on increasing with every next trophic level in a food chain. This is called as biological magnification.

Maximum concentration of such chemicals is found to accumulate in human body as human occupies the top level in any food chain.



# TOPIC-2 BIODEGRADABLE AND NON–BIODEGRADABLE SUBSTANCES

## Managing the Garbage

In our daily activities, we generate a lot of material that are thrown away. These materials are classified as -

### Biodegradable Substances :

Substances that are broken down by biological processes are said to be **biodegradable.**
Ex : fruit and vegetable peels, cotton, jute, cow-dung, paper, etc.

### Non-Biodegradable Substances :

Substances that are not broken down in this manner are said to be **non-biodegradable.**
Ex : Plastic, polythene, metals, synthetic fibres, radioactive wastes, pesticides, etc.

These substances may be inert and simply persist in the environment for a long time or may harm the various members of the eco-system.

### Methods of Waste Disposal:

- Biodegradable and non-biodegradable wastes should be discarded in two different dustbins.
- Organic wastes are filled in a compost pit and covered with a layer of soil. After about three months, garbage changes to manure.
- Biodegradable waste produced by animals can be used in biogas plant to produce biogas and manure.
- It is a conventional technique to reuse an item again e.g., newspaper for making envelops.
- Non-biodegradable waste are recycled to make new items.

## Ozone Layer

- Ozone (O3) is a molecule formed by three atoms of oxygen.
- Ozone, is a deadly poison. However, at the higher levels of the atmosphere, ozone performs an essential function. It shields the surface of the earth from ultraviolet (UV) radiation from the Sun.
- This radiation is highly damaging to organisms, for example, it is known to cause skin cancer in human beings.

## Formation of Ozone Layer

Ozone($O_3$) at the higher levels of the atmosphere is a product of UV radiation acting on oxygen ($O_2$) molecule. The higher energy UV radiations split apart some moleculer oxygen ($O_2$) into free oxygen (O) atoms. These atoms then combine with the molecular oxygen to form ozone as shown—

$$O_2 \xrightarrow{UV} O + O$$

$$O + O_2 \rightarrow O_3$$
(Ozone)

The amount of ozone in the atmosphere began to drop sharply. This decrease has been linked to synthetic chemicals like chlorofluorocarbons (CFCs) which are used as refrigerants and in fire extinguishers.

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